

BLAND NEW WORLD: IS AI MAKING US ALL THINK THE SAME?

Researchers worry that generative AI homogenizes culture and cognition. **By Matthew Hutson**



ILLUSTRATION: PAWEŁ JONCA

Lhivar Sourati often gets déjà vu while perusing recent research in his field of computer science. “I read papers and I’m like, ‘I’ve seen this paper before,’” he says. Everything’s looking kind of the same, shorn of individual quirks, says Sourati, who is a PhD student at the University of Southern California in Los Angeles.

Sourati suspects artificial intelligence is to blame. Because people are increasingly relying on large language models (LLMs), their writing is growing more similar, he suggests.

Now, Sourati and other researchers are putting their anecdotal observations to the test by exploring how generative AI (genAI) – systems that create text, images and other output – might be homogenizing both culture and cognition. In a March paper, he and his co-authors noted that this phenomenon is similar to the concept of ‘McDonaldization’, invoking how characteristics of the fast-food industry, such as efficiency and predictability, have influenced society and led to more uniformity¹.

GenAI is different from previous technologies that merely spread information, the researchers say, in that it actively shapes it. Studies suggest that AI tools can flatten language use, creativity and cultural values, and there is evidence that the technology can even influence the decisions we make, the way we act and the opinions we hold.

So how big a problem is this? And what can be done to combat it? Although AI homogenization hasn’t upended society and might never get to that point, Sourati imagines a worst-case outcome in the long term, in which humans collectively become less adaptable. “That’s actually really scary for me.”

Mind hijacking

Emily Wenger, a computer scientist at Duke University in Durham, North Carolina, says the risk of homogenization is an existential one. “If we’re all using AI to write our e-mails or whatever, what do we become as a species?”

We might fall prey to groupthink, she warns. “We have suffered as a society when we silence or ignore edge voices.”

Wenger has investigated the creative homogeneity of AI models. She and a colleague tested 22 LLMs and 102 people on three creative tasks, including one on divergent thinking that asked the models and participants to name alternative uses for common objects. The pair found² that LLMs produced ideas that were slightly more original – more semantically different from the question – than people did. But the LLMs’ responses were more similar to each other than the human responses were.

Other researchers mapped the narrative features of short stories written by people and five LLMs by analysing the plots, characters, settings and other elements³. They found that

the AI-generated stories clustered together on the basis of their narrative features, but that those in the human-written ones were more spread out.

Working with AI tools can also reduce human creativity. In a 2024 study, people wrote eight-sentence short stories on assigned topics. Some could first ask an LLM for ideas. AI use led to stories that evaluators rated as more novel and enjoyable (except for those by the most talented writers). But it also made stories more similar to each other⁴. In a 2025 study that involved brainstorming tasks, the ideas people submitted were more creative (according to human evaluators) when they used ChatGPT, but again AI use reduced the diversity of the ideas⁵. Recent meta-analyses combining several studies have supported these observations. One published in April⁶ found that AI homogenization was strongest for idea generation, especially in complex or constrained tasks.

Similar effects are showing up outside laboratory experiments. One study examined more than 400,000 scientific articles in the Web of Science database and found that after ChatGPT’s release in late 2022, the number of articles per author increased, but so did their similarity in content and linguistic style⁷.

Other research has focused on the use of language. Sourati and his colleagues studied local news articles, arXiv preprints and posts on the social-media platform Reddit and found that after ChatGPT’s initial release, linguistic measures of writing style showed a decrease in variation. The researchers reproduced this effect by using LLMs to correct the grammar in human-written texts, which erased many signifiers of personality, moral values and demographics⁸.

GenAI can also mute cultural differences. In one study⁹, people in India and the United States were asked to describe their favourite rituals, heroes and symbols, and to write an e-mail in a way that would demonstrate their values. Half wrote using an autocomplete tool that suggested up to nine words each time they paused. This use of AI increased writing similarity between Indians, between Americans and between the two groups, leading Indians to sound more American in their wording. It also led to fewer rich details about, say, the Indian festival of Diwali; instead, descriptions in the writing samples were more generic.

The output of AI models can affect the ideas of those who view it. One study before the 2024 US presidential election found that LLMs were biased towards the initial Democratic nominee, Joe Biden, over the Republican candidate Donald Trump, and that chatting with the models about politics made Trump supporters less favourable towards him¹⁰.

LLM content can corrupt the cognition of both consumers and creators. Mor Naaman, an information scientist at the research

centre Cornell Tech in New York City, who co-authored the study on the use of autocomplete tools, thinks that genAI creates both cultural and cognitive imperialism⁹. The autocomplete feature, he says, leads to “mind hijacking”, altering what users think. In one study that he worked on¹¹, participants wrote about social media with the help of an LLM that was secretly prompted to argue that social media was good or bad. The LLM affected not only what users wrote, but also their attitudes in a post-writing survey. In a follow-up study, people’s beliefs about issues such as the death penalty skewed towards that of the model’s prompt even when they were surveyed weeks later, and when they had been warned about the LLM’s bias¹².

In work by other researchers, half the participants had access to ChatGPT for five days to assist with tasks that tested their creativ-



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ity, and half did not. When all participants completed tasks two months later, none of them had AI access, but those with previous AI access still gave answers that were more similar to each other. The researchers called this a “creative scar”⁷.

Sourati describes the impact of AI on cognition by invoking George Orwell’s 1949 dystopian novel *Nineteen Eighty-four*. “The way that you talk affects the way that you reason,” Sourati says. In other words, if AI use leads everyone to talk the same, they might reason the same, too, he says.

Taking the average

The homogenization of AI output and, in turn, users’ thoughts, have causes that are, respectively, technical and psychological, experts say. When a generative model fails to provide a variety of outputs, it’s called mode collapse. This can happen for a few reasons: one is that there is limited diversity in the data used to train the models. The second is that they are rewarded for predicting likely patterns, not divergent ideas. A third is that they are fine-tuned to please human raters, who don’t usually prioritize novelty in responses.

There’s another phenomenon called model collapse. Definitions vary, but broadly it means that if a model is trained on output from itself or another model – as will increasingly happen now that AI tools are filling the Internet with slop – after a few generations, the output



The idea of 'McDonaldization', in which society becomes more uniform, also applies to AI use.

degrades or loses diversity.

Even without model collapse, AI systems can actively bias human culture to become more AI-like and homogeneous. Studies have found that LLMs give a higher rating to CVs and works that they have produced than to those written by other models or humans¹³. That could incentivize us to appease them.

Psychologically, Naaman says, people tend to follow an LLM's lead because some look to it as being either representative of others' views or a sage that knows what's best. And when they use an LLM in their work, as in the auto-complete studies, they might feel some ownership of its output. People generally adjust their attitudes to match their own self-perceived behaviour, which can lead them to adopt the perspectives of AI models.

AI-produced ideas also have an anchoring effect that fixes our thinking, says Alwin de Rooij, a creativity researcher at Tilburg University in the Netherlands.

Overall, he calls homogenization a "socio-technical" issue: "It's not really a problem of the AI, it's not really a problem of the people. It's more like an emergent property of human-AI co-creation."

Difference makers

Researchers have suggested and attempted many fixes at various stages of the pipeline. The first step would be to diversify the data that models are trained on. Another option would be to alter the way training happens. One team created Diversity-Aware Reinforcement Learning (DARLING), which rewards models for responses that are both good and different from other responses. This simultaneously improved quality and diversity¹⁴.

One can also prompt trained models differently. Researchers at the technology firm Meta

and the Massachusetts Institute of Technology in Cambridge prompted an LLM to first detect what kind of problem it was solving, such as mathematics or creative writing, then to choose an appropriate self-prompt that seeks variety. In the case of an algebra question, for example, the prompt might ask for several solution strategies. This increased diversity while maintaining quality¹⁵.

THERE'S AN ANTI-AI MOVEMENT."

Other researchers have eked diversity from models by prompting them with random inputs, or by asking them to role-play as a multicultural cast of characters. Researchers have also suggested that users prompt LLMs with socio-cultural context, or with examples of their own writing style so that it maintains their voice.

We can do more than limit AI homogenization, researchers say – we can use the technology to increase cultural and cognitive diversity. De Rooij says that although LLMs can make people fixate on an average idea, they might also divert us from our own average ideas, leading us in new directions, even if we don't use the LLMs' suggestions. And researchers have used evolutionary algorithms and techniques for open-ended exploration to propose ideas, such as new algorithms or game strategies, that humans might never have imagined.

No one can predict the long-term repercussions of AI homogenization. Although some researchers have strong concerns, others offer caveats, uncertainty and even optimism.

De Rooij, who conducted one of the meta-analyses⁶, says the effects were small on average. "So it's not like there's this complete collapse, right? It's not that suddenly everything looks the same. But then thinking about the scale of AI adoption, it is consequential." He wants more longitudinal studies of AI's homogenizing effect.

Naaman says the question of whether genAI will increase or decrease diversity is a false dichotomy. "Twenty years later, we're still debating what kind of social-media use is helpful," he says. Some people might use AI to produce slop, but others might use it to augment their expressive abilities.

Alberto Acerbi, a cognitive anthropologist at the University of Trento in Italy, is relatively hopeful about AI homogenization. Looking at the history of globalization, he says, as a result "you had fewer languages, but then you had some emerging microcultures in other places". Whether globalization reduces diversity depends on what you measure.

Overall, "it seems that cultures tend to be quite resilient about getting too homogenized", Acerbi says. That's in part because individuals often avoid conformity and seek unique niches.

That urge might be especially strong when there is a risk of being mistaken for a machine. It is becoming increasingly common, for example, for people to avoid words and phrases they associate with LLMs.

"There's an anti-AI movement," Naaman says. "We could have another future where we will have to be really creative to stand out."

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